



Review Articles

Understanding spinal cord astrocytoma: Molecular mechanism, therapy, and comprehensive management

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ARTICLE INFO

Keywords:

Spinal cord tumor
DMG
H3 K27altered
Liquid biopsy
Tumor microenvironment

ABSTRACT

Spinal cord astrocytoma is a rare and highly debilitating tumor, yet our knowledge of its clinical characteristics, molecular features, and pathogenesis remains limited compared to that of its counterparts in the brain. Current diagnostic and therapeutic approaches for spinal cord astrocytomas are primarily based on established guidelines for brain astrocytomas. However, recent studies have revealed unique clinical and pathological attributes that distinguish spinal cord astrocytomas from their corresponding brain counterparts. These findings underscore the inadequacy of directly applying the clinical guidelines developed for brain astrocytomas to spinal astrocytomas. In this review, we provided an up-to-date overview of the advancements in understanding spinal cord astrocytomas. We also discussed the challenges and future research prospects in this field with the aim of improving the precision of diagnosis and therapy for these tumors. Specifically, we emphasized the importance of enhancing our understanding of the molecular heterogeneity, immune characteristics, and clinical trials of spinal cord astrocytomas.

1. Introduction

Based on histology features, the non-ependymal glial tumors of the spinal cord are mainly spinal cord astrocytomas, which constitute 30–40 % of all intramedullary tumors and occur at a ratio of approximately 1:10 compared to their intracranial counterparts [1,2]. Although spinal cord astrocytomas can occur in patients of all ages, they are most common in young adults. Our previous studies have indicated that the median age of patients with spinal cord astrocytoma is approximately 35 years, with a higher prevalence among males [3,4]. The tumor often occurs in the cervical and thoracic segments, followed by the lumbar spinal cord [5,6]. Patients typically present with diffuse nonspecific axial pain, including back, radicular, and central pain, accompanied by gradual progressive sensory or motor disturbances [7]. Magnetic resonance imaging (MRI) is the most common method for diagnosing spinal

cord glioma before surgery and often shows edema and thickening at the location of spinal cord lesions, with a high signal on T2 weighted images and low or equal signals on T1 weighted images [8,9]. Distinguishing spinal cord astrocytomas from other spinal space-occupying diseases remains challenging owing to their atypical symptoms and variable clarity of tumor boundaries on MRI scans [8,10]. Thus, a prompt preoperative diagnosis of spinal cord astrocytoma after its occurrence is difficult. A recent clinical study showed that the average time from tumor occurrence to diagnosis is approximately 3 years [6]. Given the high rates of disability and mortality associated with spinal cord astrocytomas, novel preoperative testing methods are urgently required (Fig. 1).

Surgical resection is the primary treatment approach for spinal cord astrocytomas [11]. Surgeons rely on preoperative MRI findings and microscopic examinations to determine the extent of the resection. Intraoperative techniques, such as rapid frozen section pathology,

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<https://doi.org/10.1016/j.canlet.2024.217154>

Received 16 June 2024; Received in revised form 31 July 2024; Accepted 1 August 2024

Available online 8 August 2024

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Abbreviation		WHO	World Health Organization
CAR-T	chimeric antigen receptor T-cell	TAM	Tumor-associated macrophage
cfDNA	cell-free DNA	TCGA	The Cancer Genome Atlas
CGGA	Chinese Glioma Genome Atlas	TMZ	temozolomide;
CNS	central nervous system	ACVR1	Actin A receptor type 1
CSF	cerebrospinal fluid	ATRX	Alpha thalassemia/mental retardation, X-linked
DIPG	diffuse intrinsic pontine glioma	EGFR	Epidermal growth factor receptor
DMG	diffuse midline glioma	EZH2	Enhancer of zeste homolog 2
GBM	glioblastoma	EZH1P	EZH inhibitory protein
Gy	Gray	FGFR1	Fibroblast growth factor receptor 1
HGAP	high-grade astrocytoma with piloted features	GD2	disialoganglioside;
MRI	magnetic resonance imaging	H3	Histone 3
NCCN	National Comprehensive Cancer Network	HDAC	Histone deacetylase
PA	pilocytic astrocytoma	IDH	isocitrate dehydrogenase
PCR	polymerase chain reaction	MGMT	O6-methylguanine DNA methyltransferase
PDC	patient-derived cell	K27	lysine 27
PDX	patient-derived xenograft	PDGFRA	Platelet-derived growth factor receptor alpha
SCA	spinal cord astrocytoma	PPM1D	Protein phosphatase 1D
		TP53	Tumor Protein 53

fluorescein sodium imaging, and neuroelectrophysiological monitoring, have also been utilized to guide surgical procedures [12–16]. Radiotherapy is also adopted for aggressive tumors after surgery [17,18]. However, because of the delicate nature of the spinal cord, both surgical resection and subsequent radiotherapy pose risks of spinal cord injury and other severe side effects. Achieving optimal tumor removal while preserving the normal structure and function of the spinal cord remains challenging in the treatment of spinal cord astrocytomas (Fig. 1).

The latest 5th WHO classification of CNS tumors emphasizes the importance of considering both histological and molecular features in the post-surgery pathological diagnosis of spinal cord astrocytomas [19, 20]. Traditionally, these tumors have been categorized as pilocytic

astrocytomas or diffuse astrocytomas, ranging from grades 1 to 4, based on their histological features. The 5-year survival rates for these grades vary, with 100 % for grade 1, 77 % for grade 2, 61 % for grade 3, and 7 % for grade 4, respectively [21]. The molecular characteristics and biological aggressiveness can further stratify these tumors into various types or subtypes that may not align with their histological diagnoses [22,23]. Spinal cord gliomas featuring H3 K27M mutation should be diagnosed as “diffuse midline gliomas, H3 K27 altered, grade 4,” regardless of their histology grade [20]. Additionally, molecular features such as BRAF and H3 K27 alterations are associated with the pathogenesis of spinal cord astrocytomas and may offer potential avenues for novel therapeutic approaches (Fig. 2) [4,22,24,25].

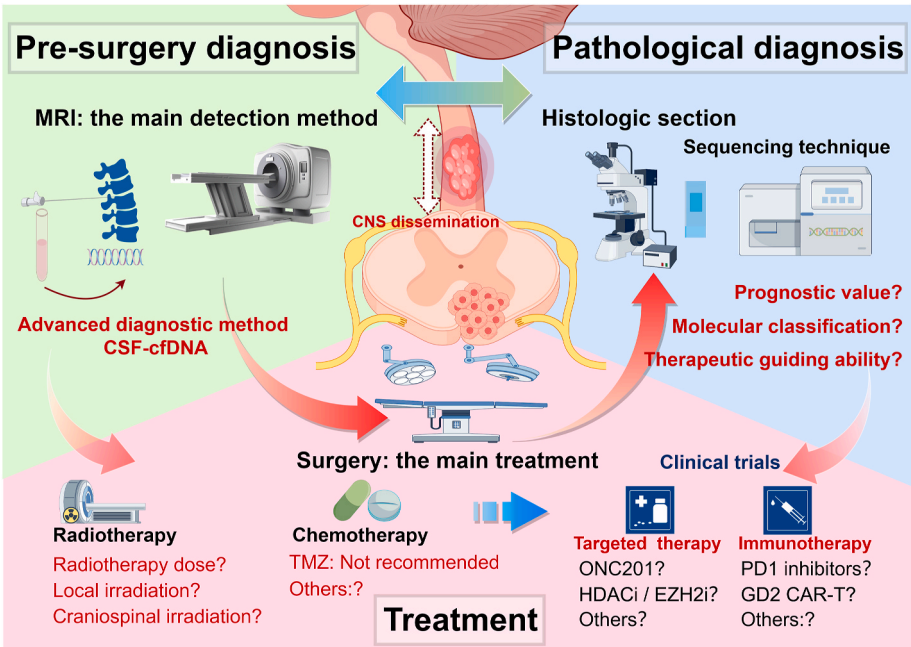


Fig. 1. Current clinical diagnosis and treatment of spinal cord astrocytomas. Diagnosis: MRI is the most important preoperative diagnostic method for spinal cord gliomas. In addition, new detection techniques such as liquid biopsies have great application prospects. Treatment: Surgery is currently the primary treatment modality, with postoperative pathological diagnosis mainly focused on histopathological and molecular pathological diagnoses. Radiotherapy constitutes the main postoperative treatment for spinal cord astrocytomas. However, the site and dose of radiotherapy remain controversial, especially for gliomas with central dissemination. Temozolomide chemotherapy is not recommended; however, effective chemotherapies, targeted therapies, and immunotherapies remain lacking.

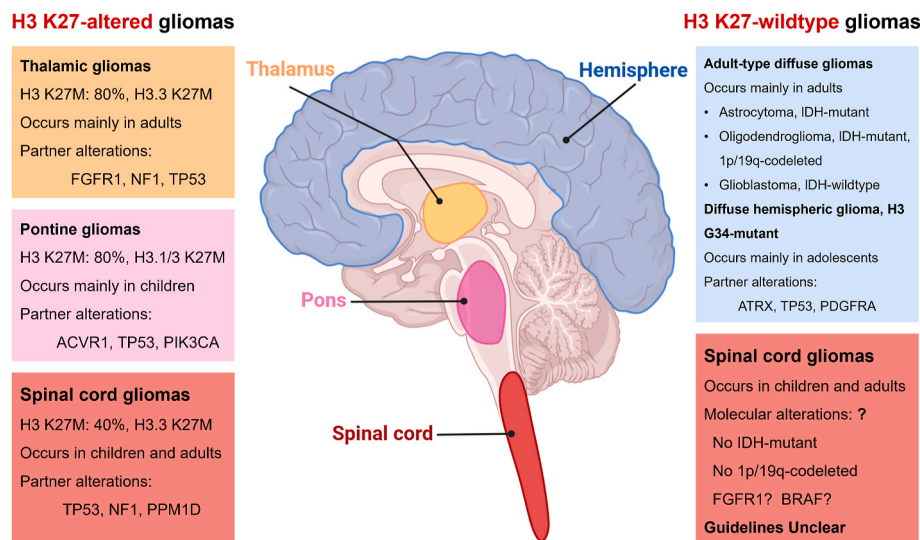


Fig. 2. Clinical and pathological features of spinal astrocytoma compared with gliomas in other parts of the central nervous system.

Gliomas can occur in the supratentorial brain, midline thalamus, pons, or spinal cord. Previous studies have demonstrated that diffuse midline gliomas, with H3 K27 altered, have distinct molecular phenotypes. H3K27M incidence was up to 80% in thalamic and pontine gliomas and around 40% in spinal cord gliomas. However, spinal cord gliomas differ from supratentorial and brainstem gliomas in terms of clinical features, such as age, prognosis, and molecular variants.

In particular, studies have reported that spinal cord gliomas exhibit a greater tendency for dissemination within the central nervous system in comparison to brain gliomas, despite the significantly higher incidence of the latter [26–29]. Recently, we performed a comprehensive descriptive study of CNS dissemination of spinal astrocytomas based on a single-center clinical cohort of 94 patients. We found that the CNS dissemination rate of SCA was as high as 85.0 % for all spinal cord gliomas at 5 years of follow-up. Notably, this ratio was even higher, reaching 95.2 %, in H3 K27M-mutant gliomas. H3K27-altered status was an independent risk factor of CNS dissemination and the dissemination occurred in all cases of H3 K27M-mutant that reached the endpoint of follow-up, suggesting that CNS dissemination has become a prevalent feature and a key determinant of lethality in patients with SCA. However, owing to the rarity of these tumors, our understanding of their pathological characteristics and biological behavior remains limited, with most knowledge derived from their brain counterparts. Consequently, achieving an integrated diagnosis of spinal cord astrocytomas based on the latest WHO classification of CNS tumors remains challenging.

Recent studies have highlighted the clinical, molecular, and micro-environmental heterogeneity among gliomas with the same histological type but different locations [30,31]. These findings underscore the significance of a precise diagnosis based on the molecular features of spinal cord astrocytomas. Additionally, advancements in molecular understanding can contribute to improved patient management through a better understanding of the underlying biology [32]. In this review, we aimed to present the latest advancements and challenges in the diagnosis, treatment, molecular features, and pathogenesis of spinal cord astrocytomas. Furthermore, we discussed potential directions for enhancing the precision of diagnosis and therapy for these rare yet aggressive tumors.

2. Advances in diagnosis, molecular features, and treatment

2.1. Preoperative diagnosis

MRI is the most common method for pre-surgical assessment of spinal cord gliomas. Typically, MRI imaging reveals edematous and thickened regions in the affected spinal cord, often asymmetrical and slightly off-center. T2 weighted images usually display high signal intensity, while T1 weighted images generally show low or equal signal.

Additionally, TI enhancement imaging typically reveals uneven enhancement of the lesion [8,9]. Traditional MRI imaging examinations often face challenges in distinguishing spinal cord astrocytomas from other intramedullary space-occupying lesions, such as inflammatory demyelinating lesions. This difficulty arises from the complex anatomical structure of the spinal cord and the varied clinical presentation of tumor growth [33,34]. Recent advancements in pre-surgery testing methods have shown potential for enhancing the diagnostic accuracy of spinal cord astrocytomas. Advanced MRI techniques, such as diffusion tensor imaging, diffusion tensor tractography, and other multimodal MRI examinations, offer improved evaluation of lesion texture and the relative relationship between the tumor and nerve fibers. These advancements aid in differential diagnosis and facilitate precise surgical planning. We previously developed a deep learning model based on preoperative multimodal magnetic resonance images to predict the H3 K27M status and prognosis of spinal cord astrocytomas before surgery (Fig. 1) [35,36].

In addition to imaging techniques, molecular testing of liquid biopsies, particularly cerebrospinal fluid (CSF), has shown promise in enhancing the pre-surgery diagnostic accuracy of spinal cord astrocytomas [37–39]. CSF-based diagnostics have been particularly effective for tumors located in close proximity to the CSF, such as brain stem tumors and disseminated medulloblastoma [37–39]. In a recent comprehensive investigation, we evaluated the performance of targeted sequencing of cell-free DNA (cfDNA) derived from CSF to differentiate intramedullary spinal cord tumors [40]. Our findings demonstrated that CSF cfDNA-targeted sequencing is highly effective in the differential diagnosis of H3 K27M-mutant spinal cord astrocytomas, with a detection rate of 85.7 % and 100 % specificity for H3 K27M mutations. However, the diagnostic utility of CSF cfDNA testing for low-grade spinal cord H3-wildtype astrocytomas has remained challenging. Detection of tumors can be challenging owing to their potential for very low levels of circulating tumor DNA in the CSF. Methods with higher sensitivity, such as droplet digital PCR or nanopore sequencing, may have the potential to detect minute quantities of circulating tumor DNA associated with spinal cord H3-wildtype astrocytomas [41,42]. Furthermore, other biomarkers, such as profiles and proteomics, hold promise for use in the differential diagnosis of spinal cord astrocytomas [43]. These approaches may provide additional information to complement the molecular testing of CSF samples and further improve diagnostic accuracy (Fig. 1).

2.2. Pathological diagnosis, molecular features, and pathogenesis

The publication of the 5th WHO classification of tumors of the central nervous system has led CNS tumors into an era of integrated diagnosis, which incorporates both histological and molecular features [19, 20]. However, challenges remain in accurately diagnosing spinal cord astrocytomas because of limited information on their molecular features [44]. Our research, along with other studies, has demonstrated that the H3 K27M mutation is only identified in 40 % of spinal cord astrocytomas, a notable lower rate compared to that of brain stem diffuse gliomas (which exhibit rates above 80 %) [4,45–47]. This disparity renders the definitive classification of H3-wildtype tumors largely unresolved (Fig. 2).

A limited number of studies have indicated that spinal cord H3-wildtype astrocytomas exhibit high molecular heterogeneity, potentially including diagnoses such as pilocytic astrocytoma (PA), glioblastoma IDH wildtype (GBM), high-grade astrocytoma with piloid features, among others [22,24,48,49]. However, recent studies have found that the prediction scores of DKFZ DNA methylation profile-based predictors, a powerful tool for the classification of CNS tumors, for spinal cord H3 K27M-wildtype astrocytomas are not yet sufficiently accurate [23,24]. This may be attributed to differences in methylation profiles between spinal cord and brain tumors. A recent study of 15 spinal cord PA discovered that spinal cord PA forms a unique methylation cluster distinct from other midline and posterior fossa PAs and that FGFR1 mutations occur in 36 % of spinal cord PA, which is higher than that in previous reports of PAs in other locations [43]. Additionally, a single-cell study has indicated that the tumor microenvironment, particularly myeloid cells, may contribute significantly to differentially expressed genes among PAs with different brain locations [50]. These findings suggest an urgent need to comprehensively characterize the molecular mechanism and microenvironment of spinal cord H3 K27M-wildtype astrocytomas to promote their precise diagnosis and understanding of the underlying pathogenesis of these tumors.

Although all midline diffuse gliomas, including tumors in the thalamus, brain stem, and spinal cord, with H3 K27M/I mutations or EZHIP overexpression, should be diagnosed as DMG, H3 K27 altered, grade 4, recent studies have revealed the heterogeneity of H3 K27-altered DMGs across various localizations [21,30,31,51–56]. In general, tumors located in the pons/brain stem are often diagnosed at younger ages and are associated with worse overall survival compared to cases located in the thalamus [51,52,56]. Tumors harboring K27M mutations in canonical H3.1 or H3.2, encoded by HIST1H3B/C or HIST2H3B genes, are enriched in the pons, accounting for approximately 20 % of H3 K27-altered cases in this location. These tumors predominantly affect younger children [30,51]. Tumors located in the thalamus primarily affect adult patients and are exclusively associated with H3.3-mutant cases. Recently, distinct cellular origins and prognostic characteristics have been revealed in H3 K27M-mutant midline gliomas of different anatomic localizations and ages [14,20–24]. Our studies have shown that majority of the patients with H3 K27M-mutant spinal cord diffuse glioma are adults, and their prognosis is better than that of patients with H3 K27M-mutant brain stem tumors, who were predominantly pediatric patients [43,57]. In addition, our study indicated that the histopathological grade of spinal cord gliomas retained significant prognostic value independent of H3K27M mutation [4,25,44]. These findings challenge the current NCCN Guidelines, which primarily recommend referring to guidelines for managing brain counterparts in the clinical management of spinal cord diffuse glioma [58,59].

In terms of pathogenesis, H3 K27 altered DMGs are primarily driven by a global loss of H3 K27 trimethylation (H3 K27me3) caused by the H3 K27M/I mutation or EZHIP overexpression. However, it should be noted that H3 K27-alterations alone are insufficient for tumorigenesis; at least one partner alteration is required for tumor transformation. These partner alterations include TP53 or PPM1D mutations, ACVR1 mutations, ATRX mutations, PDGFRA amplification, and FGFR1 or BRAF

mutations [60–62]. Furthermore, animal models have shown that H3 K27-altered tumors with different partner alterations exhibit different growth rates, as well as spreading and invasion phenotypes [61]. Interestingly, different partner mutations also correlate with varying ages, anatomical locations, and clinical outcomes in brain H3 K27-altered DMGs [4,31,51,52,61]. ACVR1 mutations also occur mainly in H3.1/3.2 K27M-mutant tumors [63]. In contrast, FGFR1 mutations tend to occur in the thalamus or other midline brain locations outside the pons in adult cases with a relatively better prognosis [51,64]. Our recent study reported targeted sequencing results of 34 spinal cord H3 K27M-mutant astrocytomas and found that 58.8 % revealed mutations in TP53, which were mostly mutually exclusive to PPM1D mutations (26 %) and NF1 mutations (44 %) [43]. We did not find BRAF or EGFR1 mutations in these tumors. Another study using whole-exome sequencing of pan-spinal cord astrocytomas did not detect BRAF or EGFR1 mutations in spinal cord H3 K27M-mutant astrocytomas. Taken together, these findings suggest that the molecular features and associated biological consequences may vary among H3 K27-altered DMGs located in different regions (Fig. 2).

2.3. Treatment

Surgery remains the primary approach for treating spinal cord astrocytomas. Intraoperative neurophysiological monitoring is increasingly utilized during intramedullary surgery to safeguard normal spinal cord function. These include monitoring techniques such as motor-evoked potentials, somatosensory-evoked potentials, electromyography, and D-wave recordings [65,66]. Numerous studies have demonstrated that intraoperative neurophysiological monitoring can expand the resection range of intramedullary tumors, enhance the preservation of patient nerve function, decrease postoperative complications, and ultimately improve prognosis [65,67]. Furthermore, neurosurgeons can develop refined plans by employing various surgical techniques. Notably, gliomas often occur on the ventrolateral side of the spinal cord, often posing challenges when utilizing traditional posterior median approaches, which may inadvertently damage the posterior cord [68,69]. To address this issue, the dorsolateral sulcus approach (spinal dorsal root entry zone spinal cordotomy) has been employed [15]. This technique enables better preservation of the contralateral nerve function while reducing the risk of damage to the ipsilateral posterior lateral bundle. As reported, the method of accessing the spinal cord through the spinal dorsal root entry zone has been proven successful in the removal of intramedullary gliomas [69–71]. This approach minimizes damage to the posterior cord and posterior horn while also reducing the risk to the spinal vessels. Consequently, it protects the patient's nerve function and offers certain benefits in terms of prognosis. Despite these advances, further refinements in spinal cord surgical techniques are still required to improve both the quality of life and overall survival of patients.

The efficacy of radiotherapy for the treatment of spinal cord gliomas is a topic of ongoing debate. This controversy arises from variations in treatment plans among different research teams, a lack of larger research cohorts, and improved patient classification. According to the NCCN guidelines, postoperative observation is recommended for PAs and for completely resected low-grade astrocytomas. However, patients with high-grade spinal cord astrocytomas, such as H3 K27-altered tumors, are advised to undergo radiotherapy 4–6 weeks after surgery [17]. Some studies have demonstrated a dose-dependent therapeutic response to radiotherapy for spinal astrocytomas [3,72]. However, considering the tolerance of the spinal cord and cauda equina nerves, the currently accepted radiotherapy dose ranges from 40 to 50.4 Gy [18]. Lower radiation levels may not be sufficient to eliminate tumors, whereas excessive doses can lead to spinal cord toxicity and other complications associated with radiotherapy. Furthermore, substantial evidence indicates that radiotherapy-induced alterations in the microenvironment, along with the development of radioresistance, significantly complicate the treatment of tumors [73,74]. Emerging techniques such as

intensity-modulated, stereotactic, dynamic conformal, and three-dimensional conformal radiotherapies have been developed to mitigate the side effects of radiotherapy through target optimization and dose segmentation [75–77]. By utilizing these techniques, radiation oncologists can optimize treatment plans and achieve better target coverage while minimizing the radiation dose to surrounding healthy tissues. This approach enhances the potential for higher-dose radiotherapy plans that are both effective and safe for patients with spinal cord gliomas; however, it still requires investigation in a large cohort to validate its efficacy and safety. Nevertheless, it is difficult to perform radiotherapy in patients with central nervous system dissemination. Craniospinal irradiation may be an option, but the safety and efficacy of radiotherapy still need to be studied further [78].

Temozolomide is a first-line chemotherapeutic drug for brain GBM. Radiotherapy combined with temozolomide chemotherapy can significantly improve the prognosis of patients with MGMT promoter-methylated glioblastoma [43,79,80]. However, current clinical trials fail to provide evidence supporting the effectiveness of temozolomide in treating spinal cord gliomas [43]. Considering that the O⁶-methylguanine DNA methyltransferase (MGMT) promoter in spinal gliomas is often in an active non-methylated state [4], achieving a definitive therapeutic effect becomes challenging. Additionally, owing to the presence of the blood-brain cerebrospinal fluid barrier, most tumor chemotherapy drugs are excluded from consideration, and there is currently no recommended chemotherapy regimen for spinal cord gliomas. Intrathecal injection of cerebrospinal fluid may be a potentially effective method to improve postoperative chemotherapy for spinal cord gliomas; however, further studies are required to evaluate its efficacy.

3. Challenges and opportunities in investigating spinal cord astrocytomas

Recent advancements have shed light on the similarities and differences between spinal cord astrocytomas and their counterparts in the brain. These findings indicate that the research and treatment approaches utilized for brain astrocytomas cannot be directly applied to spinal cord astrocytomas. Nevertheless, the successful experiences and advancements in brain astrocytoma research can serve as valuable guiding principles for future investigations of spinal cord astrocytomas. In this section, we discussed the challenges and opportunities associated with studying spinal cord astrocytomas (Fig. 3).

3.1. Enhancing the diagnostic accuracy of spinal cord astrocytomas through multi-omics analysis to reveal their molecular and biological profile

Integrated multi-omics analysis has demonstrated significant potential for improving the accurate diagnosis of CNS tumors by revealing their molecular and biological characteristics [32,81,82]. However, the molecular features of spinal cord H3-wildtype astrocytomas are still largely unclear despite differences in brain tumors with similar histological features [49,83]. Therefore, there is an urgent need to establish adequate molecular classifications for spinal cord H3-wildtype astrocytomas that would facilitate their comprehensive integrated diagnosis according to the latest WHO classifications. Furthermore, understanding the molecular characteristics of spinal astrocytomas, such as BRAF fusions and FGFR1 mutations, can provide insights into their pathogenesis and therapeutic vulnerabilities. These findings highlight the importance of molecular data for the accurate diagnosis and management of H3-wildtype spinal cord astrocytomas, although further studies with larger cohorts are required to fully address these areas.

Although the diagnosis of spinal cord H3 K27 altered astrocytoma is clear, molecular subgroups of brain H3 K27 altered DMGs have also been reported [21,44,84]. There are two subgroups of tumors with different prognoses: age at diagnosis and anatomical location—namely M-pons and M-medulla. The M-pons subgroup primarily consists of

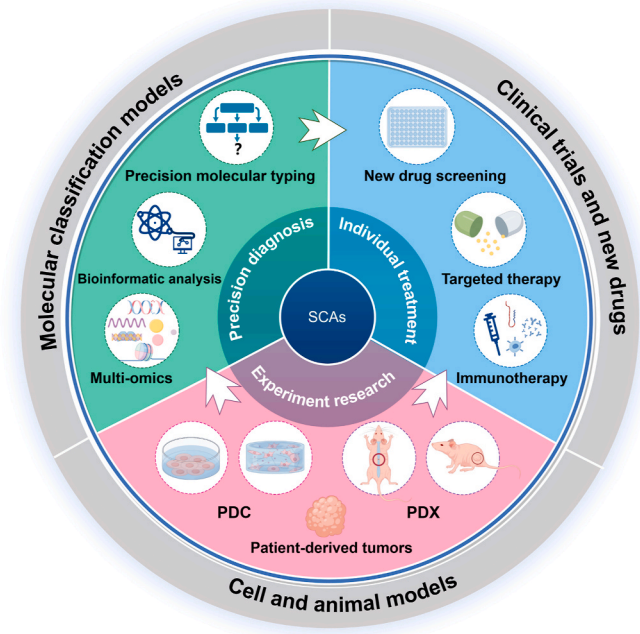


Fig. 3. Challenges for spinal cord astrocytoma research.

Currently, studying spinal cord gliomas is difficult. First, accurate diagnostic methods and pathological subtypes for guiding treatment are lacking. Therefore, it is imperative to develop a clinically useable diagnostic classification method through integrated analysis of multiple omics data. Second, there is a shortage of animal and cell models for studying spinal cord glioma treatment, highlighting the necessity to establish experimental models of spinal cord glioma. Finally, we need to accelerate spinal cord glioma microenvironment research to explore immune therapy, and emerging treatments such as targeted therapy, and promote the early transformation of clinical benefits in patients.

tumors localized in the pons, whereas most tumors in the M-medulla subgroup are localized outside the pons. Recently, a study involving a predominantly brain H3 K27-altered DMG cohort suggested that tumors carrying FGFR1 or BRAF mutations should be considered as a distinct subgroup with a relatively better prognosis [85]. Furthermore, single-cell analysis studies have shown that adult H3 K27-altered DMGs exhibit a higher proportion of mesenchymal-like tumor cells and myeloid cell infiltration than pediatric cases [31]. Spinal cord H3 K27M-mutant astrocytomas could occur in both adult and pediatric patients [4,43,44]. Additionally, our findings indicate that histological grade II/III spinal cord H3 K27M-mutant astrocytomas have significantly better survival outcomes compared to histological grade IV cases [4,44]. The latter group exhibits a higher Ki-67 index and copy number variations, suggesting heterogeneity of H3 K27 altered astrocytomas. Further studies are required to identify the molecular subgroups of spinal cord H3 K27-altered astrocytomas. These findings will not only enhance our understanding of H3 K27-altered DMGs with different anatomical locations but also improve the diagnostic accuracy of these aggressive tumors.

The establishment of databases like TCGA and CGGA has greatly advanced brain glioma research. However, these efforts have not yet extended to the development of a dedicated database for spinal cord gliomas. In the future, the construction of a comprehensive clinical and genomic map of spinal cord glioma through multi-omics technology will require the collaborative efforts of scientists worldwide [84]. We believe this will be crucial to deepen our understanding of this distinct glioma subtype and drive progress in the management of spinal cord glioma patients [86].

3.2. Establishing appropriate cell and animal models to elucidate the mechanisms and therapeutic vulnerabilities of spinal cord astrocytomas

Establishing a suitable experimental model is crucial for investigating the biological mechanisms and therapeutic vulnerabilities of tumors [87]. Currently, it is challenging to develop cell or animal models of H3-wildtype spinal cord astrocytomas, primarily because these tumors are generally less malignant than H3 K27M-altered tumors. Conversely, it is easier to establish H3 K27M-mutant cellular and animal models. Patient-derived cell (PDC) and patient-derived xenograft (PDX) models, as well as genetically engineered mouse models, have played a significant role in studying brain stem H3 K27M-mutant tumors [88,89]. These models provide valuable insights into the molecular mechanisms underlying these tumors. The H3 K27M/I mutation can reportedly result in global loss of H3 K27me3 and elevated levels of H3 K27ac by inhibiting PRC2 activity. This dysregulation of histone modifications can lead to the arrest of cell differentiation [22,90,91]. Using these models, numerous potential inhibitors have been screened for efficacy against H3 K27M-mutant tumors. These include HDAC inhibitors, EZH2 inhibitors, BET inhibitors, PDGFRA inhibitors, and others [92–96]. The identification of these inhibitors has significantly advanced our understanding of the pathogenesis of H3 K27M-mutant tumors and has paved the way for the development of novel therapeutic approaches. However, whether the progress made in understanding brain stem H3 K27M-mutant tumors can be directly extrapolated to spinal cord H3 K27-altered astrocytomas remains uncertain. As discussed earlier, spinal cord gliomas often harbor different partner mutations, such as TP53 and NF1 mutations [44]. Animal models have shown that brain H3 K27-altered tumors with different partner alterations exhibit different growth rates, spread, and invasion phenotypes, suggesting the need for tailored targeted therapeutic approaches [61]. Currently, there is only one reported PDC of spinal cord H3 K27M-mutant tumors, SU-pSCG1, which could mimic the growth of human spinal cord astrocytomas in immunodeficient mice [89]. However, the partner mutation information for this cell type has not yet been reported. Considering the heterogeneity of these cells, obtaining more PDCs with the molecular genetic characteristics of tumors is an inevitable choice for establishing cellular experimental models. Additionally, *in situ* implantation of PDC into the mouse spinal cord serves as a suitable animal model for validating *in vitro* findings. Furthermore, we can choose patient-derived tumor tissue xenotransplantation (PDX) technology to subcutaneously implant tissues from patients with spinal cord astrocytoma in mice [77]. This method is easy to perform, has a high success rate, and better reflects the proliferation and progression status of spinal cord astrocytomas *in vivo*.

3.3. Developing suitable immunotherapy approaches for spinal cord astrocytomas according to their microenvironment characteristics

Pediatric tumors with H3 K27 alterations, compared with adult brain glioblastomas, often exhibit an immune desert phenotype characterized by a lack of highly immunosuppressive and inflammatory microenvironments [97–99]. Consequently, adoptive immunotherapies such as CAR-T cell therapy targeting GD-2 and B7-H3 have been successfully used for the treatment of H3 K27M altered DMGs [100,101]. However, it is important to note that heterogeneity exists in patient responses to therapy among H3 K27-altered tumors, including differences in responses between tumors located in the spinal cord and those disseminated in the brain. These findings emphasize the significance of considering tumor heterogeneity and the specific tumor microenvironment when developing and implementing immunotherapeutic approaches for H3 K27-altered tumors. Furthermore, the immune microenvironment of H3 K27-altered tumors may be influenced by the age of the patient, as a recent study reported a higher abundance of myeloid cells in adult brain H3 K27-altered DMGs than in pediatric cases [31]. This observation aligns with recent reports demonstrating the potential benefits of H3 K27M vaccine treatment in adult H3

K27M-altered DMGs [102]. Additionally, blocking immune checkpoints, such as TIM-3, has been shown to suppress H3 K27M-altered tumors by activating the antitumor effect of macrophages and microglia [103]. Taken together, these findings suggest that immunotherapy holds promise as a potential approach for the treatment of spinal cord H3 K27M-altered DMGs. However, the development of suitable immunotherapies should consider the unique characteristics of the tumor microenvironment in these tumors. Previous studies have revealed the difference in immune microenvironment between the normal brain and spinal cord. The ratio of glial cells and neurons in the spinal cord is significantly higher than that of the brain [104]. Abundant stromal cells and blood-spinal cord barrier maintain a unique immune microenvironment different from the brain [105]. Our recent study in spinal cord ependymoma has demonstrated the presence of rich and functionally diverse immune cells in spinal ependymomas, which play a key role in tumor growth and progression. Our findings demonstrate the dual roles of CCL2⁺TAMs and CD44⁺TAMs, offering important values for precise tumor diagnosis and TAM-targeted immunotherapy strategies [106]. Ellis and colleagues respectively induced the occurrence of gliomas in the brain and spinal cord of rats and found that the microenvironment of the brain and spinal cord dominated the direction of tumor occurrence and development [107]. In addition, the different molecular pathological characteristics of H3 K27-altered gliomas in different parts of the central nervous system suggest that their microenvironment may be different. Whether previous immunotherapy methods for H3 K27-altered gliomas are suitable for spinal cord gliomas remains to be further studied. Further research is warranted to fully understand the immune landscape of H3 K27-altered tumors at different anatomical locations and to optimize immunotherapeutic strategies accordingly.

3.4. Conducting clinical trials and developing new drugs for spinal cord astrocytoma

Considering the different molecular and clinical features of spinal cord and brain tumors, the results of previous trials conducted mainly on brain tumors may not be suitable for spinal cord astrocytomas. Therefore, it is crucial to design clinical trials that specifically address the clinical characteristics, molecular genetic features, and molecular subtyping of spinal cord astrocytomas for a more precise diagnosis and treatment of patients.

In addition to the above-mentioned immune therapies, targeted therapies showing promising results in brain stem tumors with similar molecular or epigenetic changes to spinal cord tumors should be actively pursued. For example, Chamberlain et al. observed certain palliative effects of bevacizumab, an anti-angiogenic VEGF-targeted inhibitor, in a clinical cohort of six patients with spinal gliomas who exhibited continuous progression despite surgical resection, radiotherapy, and temozolomide treatment [108]. Additionally, combination therapies involving radiotherapy-sensitizing drugs have been explored to overcome the tolerance to spinal cord radiotherapy [109–111]. Furthermore, considering the challenges posed by the blood-brain cerebrospinal fluid barrier, intrathecal chemotherapy can be investigated as a potential treatment option [112]. Proteolysis Targeting Chimeras (PROTACs) represent a novel and potentially valuable therapeutic strategy for destroying “undruggable targets.” Nevertheless, numerous significant obstacles must be addressed before this can be realized in clinical treatment [113]. Currently, several clinical trials have been conducted on DIPG, including those on the HDAC inhibitor panobinostat, EZH2 inhibitor tazemetostat, and ONC201. However, preclinical studies on their applicability to spinal cord gliomas are lacking [57]. In summary, tailored clinical trials that consider these unique characteristics are needed to address the challenges associated with spinal cord astrocytomas. Exploring alternative routes of drug administration, optimizing drug combinations, and developing targeted therapies based on molecular markers or epigenetic changes are crucial avenues.

4. Summary

This review provides an overview of the current state of clinical diagnosis and treatment of spinal cord astrocytoma based on the available literature. Despite notable advancements in neurosurgical techniques, imaging, omics research, and molecular pathological testing, the precise diagnosis and effective treatment of spinal cord astrocytomas remain challenging. It is imperative to thoroughly analyze the clinical characteristics of spinal cord astrocytoma, elucidate their molecular markers, establish clinically applicable molecular subtyping methods, and develop efficient and cost-effective diagnostic technologies. Furthermore, there is a pressing need to establish robust experimental animal and cell models that offer a high level of evidence. These models will facilitate research on combined therapy, targeted therapy, and immunotherapy, which have great potential for the treatment of spinal cord astrocytomas. It is essential to acknowledge that spinal cord astrocytoma exhibits distinct clinical and pathological characteristics compared to intracranial gliomas. Therefore, it is vital to closely monitor the progress of research specific to spinal cord astrocytoma to tailor specific clinical trials, aiming to enhance the precision of clinical diagnosis and treatment and ultimately improve patient outcomes.

Funding sources

This work was supported by grants from the National Natural Science Foundation of China (82172783, 82272867), the National Key Research and Development Program of China (2019YFE0108100), and the Public Welfare Development and Reform Pilot Project of Beijing Medical Research Institute (grant no. JYY 2023–2, Tao Jiang).

The conflict-of-interest statement

All authors report no conflict of interest.

CRediT authorship contribution statement

Bo Pang: Writing – review & editing. **Songyuan An:** Writing – original draft. **Yun Liu:** Writing – original draft. **Tao Jiang:** Funding acquisition. **Wenqing Jia:** Supervision, Conceptualization. **Ruichao Chai:** Writing – review & editing, Funding acquisition. **Yongzhi Wang:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Figures were created with the help of [Figdraw.com](https://www.figma.com/).

References

- [1] M.T. Milano, M.D. Johnson, J. Sul, N.A. Mohile, D.N. Korones, P. Okunieff, K. A. Walter, Primary spinal cord glioma: a Surveillance, Epidemiology, and End Results database study, *J. Neuro Oncol.* 98 (2010) 83–92.
- [2] A.T. Parsa, J. Lee, I.F. Parney, P. Weinstein, P.C. McCormick, C. Ames, Spinal cord and intradural-extracranial spinal tumors: current best care practices and strategies, *J. Neuro Oncol.* 69 (2004) 291–318.
- [3] M.M. Abd-El-Barr, K.T. Huang, Z.B. Moses, J.B. Iorgulescu, J.H. Chi, Recent advances in intradural spinal tumors, *Neuro Oncol.* 20 (2018) 729–742.
- [4] R.C. Chai, Y.W. Zhang, Y.Q. Liu, Y.Z. Chang, B. Pang, T. Jiang, W.Q. Jia, Y. Z. Wang, The molecular characteristics of spinal cord gliomas with or without H3 K27M mutation, *Acta Neuropathol Commun* 8 (2020) 40.
- [5] F.J. Epstein, J.P. Farmer, D. Freed, Adult intramedullary astrocytomas of the spinal cord, *J. Neurosurg.* 77 (1992) 355–359.

- [6] A. Raco, V. Esposito, J. Lenzi, M. Piccirilli, R. Delfini, G. Cantore, Long-term follow-up of intramedullary spinal cord tumors: a series of 202 cases, *Neurosurgery* 56 (2005).
- [7] M. Abdel-Wahab, B. Etuk, J. Palermo, H. Shirato, J. Kresl, O. Yapici, G. Walker, B.W. Scheithauer, E. Shaw, C. Lee, W. Curran, T. Thomas, A. Markoe, Spinal cord gliomas: a multi-institutional retrospective analysis, *Int. J. Radiat. Oncol. Biol. Phys.* 64 (2006) 1060–1071.
- [8] C. Campello, E. Tabouret, O. Chinot, Challenges in diagnosis and management of adult spinal cord gliomas, *Rev. Neurol. (Paris)* 177 (2021) 515–523.
- [9] M.K. Tobin, J.R. Geraghty, H.H. Engelhard, A.A. Linninger, A.I. Mehta, Intramedullary spinal cord tumors: a review of current and future treatment strategies, *Neurosurg. Focus* 39 (2015) E14.
- [10] H. Li, Y. Weng, D. Zhou, L. Nong, N. Xu, Experience of operative treatment in 27 patients with intraspinal neurilemmoma, *Oncol. Lett.* 14 (2017) 4817–4821.
- [11] J. Klekamp, Treatment of intramedullary tumors: analysis of surgical morbidity and long-term results, *J. Neurosurg. Spine* 19 (2013) 12–26.
- [12] E. Maj, B. Szeplińska, W. Szeszkowski, M. Prokopenko, A. Cieszanowski, A. Marchel, O. Rowiński, Role of diffusion tensor imaging parameters in the characterization and differentiation of infiltrating and non-infiltrating spinal cord tumors: preliminary study, *Clin. Neuroradiol.* 30 (2020) 739–747.
- [13] Y.I. Won, Y. Choi, W.T. Yuh, S.W. Kwon, C.H. Kim, S.H. Yang, C.K. Chung, Validity of magnetic resonance imaging (MRI) in the primary spinal cord tumors in routine clinical setting, *Sci. Rep.* 12 (2022) 10151.
- [14] F. Prada, I.G. Vetrano, A. Filippini, M. Del Bene, A. Perin, C. Casali, F. Legnani, M. Saini, F. DiMeco, Intraoperative ultrasound in spinal tumor surgery, *J. Ultrasound* 17 (2014) 195–202.
- [15] A.M. Hersh, G.I. Jallo, N. Shimony, Surgical approaches to intramedullary spinal cord astrocytomas in the age of genomics, *Front. Oncol.* 12 (2022) 982089.
- [16] Z. Sun, L. Jing, Y. Fan, H. Zhang, L. Chen, G. Wang, H.S. Sharma, J. Wang, Fluorescein-guided surgery for spinal gliomas: analysis of 220 consecutive cases, *Int. Rev. Neurobiol.* 151 (2020) 139–154.
- [17] L.B. Nabors, J. Portnow, M. Ammirati, J. Baehring, H. Brem, N. Butowski, R. A. Fenstermaker, P. Forsyth, J. Hattangadi-Gluth, M. Holdhoff, S. Howard, L. Junck, T. Kaley, P. Kumthekar, J.S. Loeffler, P.L. Moots, M.M. Mrugala, S. Nagpal, M. Pandey, I. Parney, K. Peters, V.K. Puduvalli, J. Ragsdale, J. Rockhill, L. Rogers, C. Rusthoven, N. Shonka, D.C. Shrieve, A.K. Sills, L.J. Swinnen, C. Tsien, S. Weiss, P.Y. Wen, N. Willmarth, M.A. Bergman, A. Engh, NCCN guidelines insights: central nervous system cancers, version 1.2017, *J. Natl. Compr. Cancer Netw.* 15 (2017) 1331–1345.
- [18] R.D. Tendulkar, A.S. Pai Panandiker, S. Wu, L.E. Kun, A. Broniscer, R.A. Sanford, T.E. Merchant, Irradiation of pediatric high-grade spinal cord tumors, *Int. J. Radiat. Oncol. Biol. Phys.* 78 (2010) 1451–1456.
- [19] R. Chai, S. Fang, B. Pang, Y. Liu, Y. Wang, W. Zhang, T. Jiang, Molecular pathology and clinical implications of diffuse glioma, *Chin. Med. J.* 135 (2022) 2914–2925.
- [20] D.N. Louis, A. Perry, P. Wesseling, D.J. Brat, I.A. Cree, D. Figarella-Branger, C. Hawkins, H.K. Ng, S.M. Pfister, G. Reifenberger, R. Soffietti, A. von Deimling, D.W. Ellison, The 2021 WHO classification of tumors of the central nervous system: a summary, *Neuro Oncol.* 23 (2021) 1231–1251.
- [21] L.H. Chen, C. Pan, B.H. Diplas, C. Xu, L.J. Hansen, Y. Wu, X. Chen, Y. Geng, T. Sun, Y. Sun, P. Zhang, Z. Wu, J. Zhang, D. Li, Y. Zhang, W. Wu, Y. Wang, G. Li, J. Yang, X. Wang, C. Xu, S. Wang, M.S. Waitkus, Y. He, R.E. McLendon, D. M. Ashley, H. Yan, L. Zhang, The integrated genomic and epigenomic landscape of brainstem glioma, *Nat. Commun.* 11 (2020) 3077.
- [22] A. Biczok, F.L. Strubing, J.M. Eder, R. Egensperger, O. Schnell, S. Zausinger, J. E. Neumann, J. Herms, J.C. Tonn, M.M. Dorostkar, Molecular diagnostics helps to identify distinct subgroups of spinal astrocytomas, *Acta Neuropathol Commun* 9 (2021) 119.
- [23] L. Lebrun, M. Bizet, B. Melendez, B. Alexiou, L. Absil, C. Van Campenhout, N. D'Haene, S. Rorive, F. Fuks, C. Decaestecker, I. Salmon, Analyses of DNA methylation profiling in the diagnosis of intramedullary astrocytomas, *J. Neuropathol. Exp. Neurol.* 80 (2021) 663–673.
- [24] A. Misove, A. Vicha, P. Broz, K. Vanova, D. Sumerauer, L. Stolova, L. Sramkova, M. Kobizek, J. Zamecnik, M. Kyncl, Z. Holubova, P. Liby, J. Taborsky, V. Benes 3rd, I. Pernikova, D.T.W. Jones, M. Sill, T. Stancokova, L. Krskova, M. Zapotocky, Integrated genomic analysis reveals actionable targets in pediatric spinal cord low-grade gliomas, *Acta Neuropathol Commun* 10 (2022) 143.
- [25] Y.-W. Zhang, R.-C. Chai, R. Cao, W.-J. Jiang, W.-H. Liu, Y.-L. Xu, J. Yang, Y.-Z. Wang, W.-Q. Jia, Clinicopathological characteristics and survival of spinal cord astrocytomas, *Cancer Med.* 9 (2020) 6996–7006.
- [26] Z. Kong, W. Chen, D. Zhao, Y. Wang, W. Ma, Clonal evolution and supratentorial dissemination of a spinal cord glioma, *Neurol. Sci.* 42 (2021) 2137–2141.
- [27] J. Wang, L. Yi, Q.-M. Kang, J. Zhou, T.-Q. Chen, J.-P. Hugnot, S.-C. Yu, Glioma invasion along white matter tracts: a dilemma for neurosurgeons, *Cancer Lett.* 526 (2022) 103–111.
- [28] B.M. Andersen, C. Miranda, V. Hatzoglou, L.M. DeAngelis, A.M. Miller, Leptomeningeal metastases in glioma: the memorial sloan kettering cancer center experience, *Neurology* 92 (2019) e2483–e2491.
- [29] R. Babu, I.O. Karikari, T.R. Owens, C.A. Bagley, Spinal cord astrocytomas: a modern 20-year experience at a single institution, *Spine* 39 (2014) 533–540.
- [30] S. Jessa, A. Mohammadnia, A.S. Harutyunyan, M. Hulsmit, S. Varadharajan, H. Lakkis, N. Kabir, Z. Bashardanesh, S. Hebert, D. Faury, M.C. Vladiou, S. Worme, M. Coutelier, B. Krug, A. Faria Andrade, M. Pathania, A. Bajic, A. G. Weil, B. Ellezam, J. Atkinson, R.W.R. Dudley, J.P. Farmer, S. Perreault, B. A. Garcia, V. Larouche, M. Blanchette, L. Garzia, A. Bhaduri, K.L. Ligon, P. Bandopadhyay, M.D. Taylor, S.C. Mack, N. Jabado, C.L. Kleinman, K27M in

- canonical and noncanonical H3 variants occurs in distinct oligodendroglial cell lineages in brain midline gliomas, *Nat. Genet.* 54 (2022) 1865–1880.
- [31] I. Liu, L. Jiang, E.R. Samuelsson, S. Marco Salas, A. Beck, O.A. Hack, D. Jeong, M. L. Shaw, B. Englinger, J. LaBelle, H.M. Mire, S. Madlener, L. Mayr, M.A. Quezada, M. Trissal, E. Panditharatna, K.J. Ernst, J. Vogelzang, T.A. Gatesman, M. E. Halbert, H. Palova, P. Pokorna, J. Sterba, O. Slaby, R. Geyeregger, A. Diaz, I. J. Findlay, M.D. Dun, A. Resnick, M.L. Suva, D.T.W. Jones, S. Agnihotri, J. Svedlund, C. Koschmann, C. Haberler, T. Czech, I. Slavc, J.A. Cotter, K.L. Ligon, S. Alexandrescu, W.K.A. Yung, I. Arrillaga-Romany, J. Gojo, M. Monje, M. Nilsson, M.G. Filbin, The landscape of tumor cell states and spatial organization in H3-K27M mutant diffuse midline glioma across age and location, *Nat. Genet.* 54 (2022) 1881–1894.
- [32] F. Nassiri, J. Liu, V. Patil, Y. Mamatjan, J.Z. Wang, R. Hugh-White, A.M. Macklin, S. Khan, O. Singh, S. Karimi, R.I. Corona, L.Y. Liu, C.Y. Chen, A. Chakravarthy, Q. Wei, B. Mehani, S. Suppliah, A. Gao, A.M. Workewych, G. Tabatabai, P. C. Boutros, G.D. Bader, D.D. de Carvalho, T. Kissinger, K. Aldape, G. Zadeh, A clinically applicable integrative molecular classification of meningiomas, *Nature* 597 (2021) 119–125.
- [33] D.H. Kim, J.H. Kim, S.H. Choi, C.H. Sohn, T.J. Yun, C.H. Kim, K.H. Chang, Differentiation between intramedullary spinal ependymoma and astrocytoma: comparative MRI analysis, *Clin. Radiol.* 69 (2014) 29–35.
- [34] Z. Zhuo, Y. Duan, L. Qu, C. Feng, X. Huang, D. Cheng, X. Xu, T. Sun, Z. Li, X. Guo, X. Gong, Y. Wang, W. Jia, D. Tian, X. Zhang, F. Shi, S. Haller, F. Barkhof, C. Ye, Y. Liu, Automated classification of intramedullary spinal cord tumors and inflammatory demyelinating lesions using deep learning, *Radiol Artif Intell* 4 (2022) e210292.
- [35] T. Sun, Y. Wang, X. Liu, Z. Li, J. Zhang, J. Lu, L. Qu, H. Sven, Y. Duan, Z. Zhuo, D. Cheng, X. Xu, W. Jia, Y. Liu, Deep learning based on preoperative magnetic resonance (MR) images improves the predictive power of survival models in primary spinal cord astrocytomas, *Neuro, Oncol* 25 (6) (2023) 1157–1165.
- [36] J. Li, P. Zhang, L. Qu, T. Sun, Y. Duan, M. Wu, J. Wang, Z. Li, X. Gong, X. Liu, Y. Wang, W. Jia, X. Su, Q. Yue, J. Li, Z. Zhang, F. Barkhof, R.Y. Huang, K. Chang, H. Sair, C. Ye, L. Zhang, Z. Zhuo, Y. Liu, Deep learning for noninvasive assessment of H3 K27M mutation status in diffuse midline gliomas using MR imaging, *J. Magn. Reson. Imag.* 58 (3) (2023) 850–861.
- [37] C. Pan, B.H. Diplas, X. Chen, Y. Wu, X. Xiao, L. Jiang, Y. Geng, C. Xu, Y. Sun, P. Zhang, W. Wu, Y. Wang, Z. Wu, J. Zhang, Y. Jiao, H. Yan, L. Zhang, Molecular profiling of tumors of the brainstem by sequencing of CSF-derived circulating tumor DNA, *Acta Neuropathol.* 137 (2019) 297–306.
- [38] M. Lodrini, J. Graef, T.M. Thole-Kliesch, K. Astrahantseff, A. Sprüssel, M. Grimaldi, C. Peitz, R.B. Linke, J.F. Hollander, E. Lankes, A. Künkele, L. Oevermann, G. Schwabe, J. Fuchs, A. Szymanski, J.H. Schulte, P. Hundsdoerfer, C. Eckert, H. Amthauer, A. Eggert, H.E. Deubzer, Targeted analysis of cell-free circulating tumor DNA is suitable for early relapse and actionable target detection in patients with neuroblastoma, *Clin. Cancer Res.* 28 (2022) 1809–1820.
- [39] A.K. Afflerbach, C. Rohrandt, B. Brändl, M. Sönsken, J. Hench, S. Frank, D. Börnigen, M. Alawi, M. Mynarek, B. Winkler, F. Ricklefs, M. Synowitz, L. Dührsen, S. Rutkowski, A.K. Wefers, F.J. Müller, M. Schoof, U. Schüller, Classification of brain tumors by nanopore sequencing of cell-free DNA from cerebrospinal fluid, *Clin. Chem.* 70 (1) (2024) 250–260.
- [40] R. Chai, S. An, H. Lin, B. Pang, H. Yan, Y. Liu, Y. Wu, L. Wang, X. Liu, H. Chen, X. Yang, Q. Chang, W. Jia, Y. Wang, Sequencing of cerebrospinal fluid cell-free DNA facilitated early differential diagnosis of intramedullary spinal cord tumors, *npj Precis. Oncol.* 8 (2024) 43.
- [41] S. Stallard, M.G. Savelieff, K. Wierzbicki, B. Mullan, Z. Miklja, A. Bruzek, T. Garcia, R. Siada, B. Anderson, B.H. Singer, R. Hashizume, A.M. Carcaboso, K. Q. McMurray, J. Heth, K. Muraszko, P.L. Robertson, R. Mody, S. Venneti, H. Garton, C. Koschmann, CSF H3F3A K27M circulating tumor DNA copy number quantifies tumor growth and in vitro treatment response, *Acta Neuropathol Commun* 6 (2018) 80.
- [42] A.K. Bruzek, K. Ravi, A. Muruganand, J. Wadden, C.M. Babila, E. Cantor, L. Tunkle, K. Wierzbicki, S. Stallard, R.P. Dickson, I. Wolfe, R. Mody, J. Schwartz, A. Franson, P.L. Robertson, K.M. Muraszko, C.O. Maher, H.J.L. Garton, T. Qin, C. Koschmann, Electronic DNA analysis of CSF cell-free tumor DNA to quantify multi-gene molecular response in pediatric high-grade glioma, *Clin. Cancer Res.* 26 (2020) 6266–6276.
- [43] A.K. Mattox, H. Yan, C. Bettgeowda, The potential of cerebrospinal fluid-based liquid biopsy approaches in CNS tumors, *Neuro Oncol.* 21 (2019) 1509–1518.
- [44] R.C. Chai, H. Yan, S.Y. An, B. Pang, H.Y. Chen, Q.H. Mu, K.N. Zhang, Y.W. Zhang, Y.Q. Liu, X. Liu, Z. Zhao, T. Jiang, Y.Z. Wang, W.Q. Jia, Genomic profiling and prognostic factors of H3 K27M-mutant spinal cord diffuse glioma, *Brain Pathol.* 33 (2023) e13153.
- [45] M.M. Abd-El-Barr, K.T. Huang, Z.B. Moses, J.B. Iorgulescu, J.H. Chi, Recent advances in intradural spinal tumors, *Neuro Oncol.* 20 (2018) 729–742.
- [46] B. Pang, R.C. Chai, Y.W. Zhang, Y.Z. Chang, W.H. Liu, W.Q. Jia, Y.Z. Wang, A comprehensive model including preoperative peripheral blood inflammatory markers for prediction of the prognosis of diffuse spinal cord astrocytoma following surgery, *Eur. Spine J.* 30 (2021) 2857–2866.
- [47] Y. Nagashima, Y. Nishimura, K. Eguchi, J. Yamaguchi, S. Haimoto, F. Ohka, M. Takayasu, R. Saito, Recent molecular and genetic findings in intramedullary spinal cord tumors, *Neurospine* 19 (2022) 262–271.
- [48] M.Y. Deng, M. Sill, J. Chiang, J. Schittenhelm, M. Ebinger, M.U. Schuhmann, C. M. Monoranu, T. Milde, A. Wittmann, C. Hartmann, C. Sommer, W. Paulus, J. Gartner, W. Bruck, T. Rudiger, A. Leipold, Z. Jaunmuktane, S. Brandner, F. Giangaspero, P. Nozza, J. Mora, A. Morales la Madrid, O. Cruz Martinez, J. R. Hansford, T. Pietsch, A. Tietze, P. Hernaiz-Driever, I. Stoler, D. Capper, A. Korshunov, D.W. Ellison, A. von Deimling, S.M. Pfister, F. Sahm, D.T.W. Jones, Molecularly defined diffuse leptomeningeal glioneuronal tumor (DLGNT) comprises two subgroups with distinct clinical and genetic features, *Acta Neuropathol.* 136 (2018) 239–253.
- [49] A. Metais, Y. Bouchoucha, T. Kergrohen, V. Dangouloff-Ros, X. Maynadier, Y. Ajilil, M. Carton, W. Yacoub, R. Saffroy, D. Figarella-Branger, E. Uro-Coste, A. Sevely, D. Larrieu-Ciron, M. Faisant, M.C. Machet, E. Wahler, A. Roux, S. Benichi, K. Beccaria, T. Blauwblomme, N. Boddart, F. Chretien, F. Doz, C. Dufour, J. Grill, M.A. Debily, P. Varlet, A. Tauziède-Espariat, Pediatric spinal pilocytic astrocytomas form a distinct epigenetic subclass from pilocytic astrocytomas of other locations and diffuse leptomeningeal glioneuronal tumours, *Acta Neuropathol.* 145 (2023) 83–95.
- [50] Z.J. Reitman, B.R. Paoletta, G. Berghold, K. Pelton, S. Becker, R. Jones, C. E. Sinai, H. Malkin, Y. Huang, L. Grimmer, Z.T. Herbert, Y. Sun, J.L. Weatherbee, J.A. Alberta, J.F. Daley, O. Rozenblatt-Rosen, A.L. Condurat, K. Qian, P. Khadka, R.A. Segal, D. Haas-Kogan, M.G. Filbin, M.L. Suva, A. Regev, C.D. Stiles, M. W. Kieran, L. Goumnerova, K.L. Ligon, A.K. Shalek, P. Bandopadhyay, R. Beroukhi, Mitogenic and progenitor gene programmes in single pilocytic astrocytoma cells, *Nat. Commun.* 10 (2019) 3731.
- [51] A. Mackay, A. Burford, D. Carvalho, E. Izquierdo, J. Fazal-Salom, K.R. Taylor, L. Bjerke, M. Clarke, M. Vinci, M. Nandhabalan, S. Temelso, S. Popov, V. Molinari, P. Raman, A.J. Waanders, H.J. Han, S. Gupta, L. Marshall, S. Zacharoulis, S. Vaidya, H.C. Mandeville, L.R. Bridges, A.J. Martin, S. Al-Sarraj, C. Chandler, H.K. Ng, X. Li, K. Mu, S. Trabelsi, D.H. Ibrahim, A.N. Kisljakov, D. M. Konovalov, A.S. Moore, A.M. Carcaboso, M. Sunol, C. de Torres, O. Cruz, J. Mora, L.I. Shats, J.N. Stavale, L.T. Bidinotto, R.M. Reis, N. Entz-Werle, M. Farrell, J. Cryan, D. Crimmins, J. Caird, J. Pears, M. Monje, M.A. Debily, D. Castel, J. Grill, C. Hawkins, H. Nikbakht, N. Jabado, S.J. Baker, S.M. Pfister, D. T.W. Jones, M. Fouladi, A.O. von Bueren, M. Baudis, A. Resnick, C. Jones, Integrated molecular meta-analysis of 1,000 pediatric high-grade and diffuse intrinsic pontine glioma, *Cancer Cell* 32 (2017) 520–537 e525.
- [52] D. Meyronet, M. Esteban-Mader, C. Bonnet, M.O. Joly, E. Uro-Coste, A. Amiel-Benoiaich, F. Forest, C. Rousselot-Denis, F. Burel-Vandenbos, V. Bourg, J. Guyotat, F. Fenuil, A. Jouvot, J. Honnorat, F. Ducray, Characteristics of H3 K27M-mutant gliomas in adults, *Neuro Oncol.* 19 (2017) 1127–1134.
- [53] D. Castel, C. Philippe, T. Kergrohen, M. Sill, J. Merlevede, E. Barret, S. Puget, C. Sainte-Rose, C.M. Kramm, C. Jones, P. Varlet, S.M. Pfister, J. Grill, D.T. W. Jones, M.A. Debily, Transcriptomic and epigenetic profiling of ‘diffuse midline gliomas H3 K27M-mutant’ discriminate two subgroups based on the type of histone H3 mutated and not supratentorial or infratentorial location, *Acta Neuropathol Commun* 6 (2018) 117.
- [54] M. Vinci, A. Burford, V. Molinari, K. Kessler, S. Popov, M. Clarke, K.R. Taylor, H. N. Pemberton, C.J. Lord, A. Gutteridge, T. Forshe, D. Carvalho, L.V. Marshall, E. Y. Qin, W.J. Ingram, A.S. Moore, H.K. Ng, S. Trabelsi, D. H’Mida-Ben Ibrahim, N. Entz-Werle, S. Zacharoulis, S. Vaidya, H.C. Mandeville, L.R. Bridges, A. J. Martin, S. Al-Sarraj, C. Chandler, M. Sunol, J. Mora, C. de Torres, O. Cruz, A. M. Carcaboso, M. Monje, A. Mackay, C. Jones, Functional diversity and cooperativity between subclonal populations of pediatric glioblastoma and diffuse intrinsic pontine glioma cells, *Nat. Med.* 24 (2018) 1204–1215.
- [55] S. Nagaraja, M.A. Quezada, S.M. Gillespie, M. Arzt, J.J. Lennon, P.J. Woo, V. Hovestadt, M. Kambhampati, M.G. Filbin, M.L. Suva, J. Nazarian, M. Monje, Histone variant and cell context determine H3K27M reprogramming of the enhancer landscape and oncogenic state, *Mol. Cell* 76 (2019) 965–980 e912.
- [56] H.G. Vuong, T.N.M. Ngo, H.T. Le, I.F. Dunn, The prognostic significance of HIST1H3B/C and H3F3A K27M mutations in diffuse midline gliomas is influenced by patient age, *J. Neuro Oncol.* 158 (2022) 405–412.
- [57] L.J. Weisbrod, A. Thiraviyam, R. Vengoji, N. Shonka, M. Jain, W. Ho, S.K. Batra, A. Salehi, Diffuse intrinsic pontine glioma (DIPG): a review of current and emerging treatment strategies, *Cancer Lett.* 590 (2024) 216876.
- [58] B. Nabors, J. Portnow, J. Hattangadi-Gluth, C. Horbinski, NCCN CNS tumor guidelines update for 2023, *Neuro Oncol.* 25 (2023) 2114–2116.
- [59] C. Horbinski, L.B. Nabors, J. Portnow, J. Baehring, A. Bhatia, O. Bloch, S. Brem, N. Butowski, D.M. Cannon, S. Chao, M.G. Chheda, A.J. Bhatia, P. Forsyth, P. Giglio, J. Hattangadi-Gluth, M. Holdhoff, L. Junck, T. Kaley, R. Merrell, M. M. Mrugala, S. Nagpal, L.A. Nedzi, K. Nevel, P.L. Nghiemphu, I. Parney, T. R. Patel, K. Peters, V.K. Puduvalli, J. Rockhill, C. Rusthoven, N. Shonka, L. J. Swinnen, S. Weiss, P.Y. Wen, N.E. Willmarth, M.A. Bergman, S. Darlow, NCCN Guidelines® insights: central nervous system cancers, version 2.2022, *J. Natl. Compr. Cancer Netw.* 21 (2023) 12–20.
- [60] M. Pathania, N. De Jay, N. Maestro, A.S. Harutyunyan, J. Nitarska, P. Pahlavan, S. Henderson, L.G. Mikael, A. Richard-Londt, Y. Zhang, J.R. Costa, S. Hebert, S. Khazaei, N.S. Ibrahim, J. Herrero, A. Riccio, S. Albrecht, R. Kettler, S. Brandner, C.L. Kleinman, N. Jabado, P. Salomoni, H3.3(K27M) cooperates with Trp 53 loss and PDGFRA gain in mouse embryonic neural progenitor cells to induce invasive high-grade gliomas, *Cancer Cell* 32 (2017) 684–700 e689.
- [61] M. McNicholas, A. De Cola, Z. Bashardanesh, A. Foss, C.B. Lloyd, S. Hebert, D. Faury, A.F. Andrade, N. Jabado, C.L. Kleinman, M. Pathania, A compendium of syngeneic, transplantable pediatric high-grade glioma models reveals subtype-specific therapeutic vulnerabilities, *Cancer Discov.* 13 (2023) 1592–1615.
- [62] L.A. Parsels, D.R. Wahl, C. Koschmann, M.A. Morgan, Q. Zhang, Developing H3K27M mutant selective radiosensitization strategies in diffuse intrinsic pontine glioma, *Neoplasia* 37 (2023) 100881.
- [63] G. Wu, A.K. Diaz, B.S. Paugh, S.L. Rankin, B. Ju, Y. Li, X. Zhu, C. Qu, X. Chen, J. Zhang, J. Easton, M. Edmonson, X. Ma, C. Lu, P. Nagahawatte, E. Hedlund, M. Rusch, S. Pounds, T. Lin, A. Onar-Thomas, R. Huether, R. Kriwacki, M. Parker, P. Gupta, J. Becksfors, L. Wei, H.L. Mulder, K. Boggs, B. Vadodaria, D. Yergeau, J.

- C. Russell, K. Ochoa, R.S. Fulton, L.L. Fulton, C. Jones, F.A. Boop, A. Broniscer, C. Wetmore, A. Gajjar, L. Ding, E.R. Mardis, R.K. Wilson, M.R. Taylor, J. R. Downing, D.W. Ellison, J. Zhang, S.J. Baker, The genomic landscape of diffuse intrinsic pontine glioma and pediatric non-brainstem high-grade glioma, *Nat. Genet.* 46 (2014) 444–450.
- [64] E.A. Williams, P.K. Brastianos, H. Wakimoto, A. Zolal, M.G. Filbin, D.P. Cahill, S. Santagata, T.A. Juratli, A comprehensive genomic study of 390 H3F3A-mutant pediatric and adult diffuse high-grade gliomas, *CNS WHO grade 4, Acta Neuropathol.* 146 (2023) 515–525.
- [65] G. Kimchi, N. Knoller, A. Korn, Y. Eyal-Mazuz, Y. Sapir, A. Peled, R. Harel, Delayed variations in the diagnostic accuracy of intraoperative neuromonitoring in the resection of intramedullary spinal cord tumors, *Neurosurg. Focus* 50 (2021) E21.
- [66] C. Dauleac, H.-A. Leroy, R. Assaker, Minimally invasive approach for dorsal arachnoid web: 2-dimensional operative video, *Oper Neurosurg (Hagerstown)* 21 (2021) E549–E550.
- [67] C. Dauleac, S. Boulogne, C.Y. Barrey, J. Guyotat, E. Jouanneau, P. Mertens, M. Berhouma, J. Jung, N. André-Obadia, Predictors of functional outcome after spinal cord surgery: relevance of intraoperative neurophysiological monitoring combined with preoperative neurophysiological and MRI assessments, *Neurophysiol. Clin.* 52 (2022) 242–251.
- [68] C. Dauleac, I. Pellissou-Guyotat, How I do it: dorsolateral approach for ventrolateral intramedullary cavernoma, *Acta Neurochir.* 162 (2020) 1127–1129.
- [69] J. Ren, C. He, T. Hong, X. Li, Y. Ma, J. Yu, F. Ling, H. Zhang, Anterior to dorsal root entry zone myelotomy (ADREZotomy): a new surgical approach for the treatment of ventrolateral deep intramedullary spinal cord cavernous malformations, *Spine* 43 (2018) E1024–E1032.
- [70] T. Xie, Y. Feng, J. Song, G. Yao, Anterior dorsal root entry zone approach in the treatment of spinal intramedullary glioma, *World Neurosurg* 153 (2021) 139.
- [71] O. Persson, A. Fletcher-Sandersjö, G. Burström, E. Edström, A. Elmi-terander, Surgical treatment of intra- and juxtamedullary spinal cord tumors: a population based observational cohort study, *Front. Neurol.* 10 (2019) 814.
- [72] N. Nagoshi, O. Tsuji, S. Suzuki, S. Nori, M. Yagi, E. Okada, H. Okita, N. Fujita, K. Ishii, M. Matsumoto, M. Nakamura, K. Watanabe, Clinical outcomes and a therapeutic indication of intramedullary spinal cord astrocytoma, *Spinal Cord* 60 (2022) 216–222.
- [73] J. Peng, X. Yin, W. Yun, X. Meng, Z. Huang, Radiotherapy-induced tumor physical microenvironment remodeling to overcome immunotherapy resistance, *Cancer Lett.* 559 (2023) 216108.
- [74] M. De Martino, C. Daviaud, H.E. Minns, A. Lazarian, A. Wacker, A.P. Costa, N. Attarwala, Q. Chen, S.-W. Choi, R. Rabadan, L.B.J. McIntire, R.D. Gartrell, J. M. Kelly, E.C. Laiakis, C. Vanpouille-Box, Radiation therapy promotes unsaturated fatty acids to maintain survival of glioblastoma, *Cancer Lett.* 570 (2023) 216329.
- [75] J. Kahn, J.S. Loeffler, A. Niemierko, E.A. Chioocca, T. Batchelor, A. Chakravarti, Long-term outcomes of patients with spinal cord gliomas treated by modern conformal radiation techniques, *Int. J. Radiat. Oncol. Biol. Phys.* 81 (2011) 232–238.
- [76] N. Dea, Z. Gokaslan, D. Choi, C. Fisher, Spine oncology - primary spine tumors, *Neurosurgery* 80 (2017) S124–S130.
- [77] L.A. Parsels, D.R. Wahl, C. Koschmann, M.A. Morgan, Q. Zhang, Developing H3K27M mutant selective radiosensitization strategies in diffuse intrinsic pontine glioma, *Neoplasia* 37 (2023) 100881.
- [78] M. Ebad, M. Morse, T. Gooley, R. Ermoian, L.M. Halasz, S.S. Lo, J.T. Yang, M. H. Blau, M.-E. Percival, R.D. Cassaday, J. Graber, L.P. Taylor, V. Venur, Y. D. Tseng, Craniospinal irradiation for CNS leukemia: rates of response and durability of CNS control, *J. Neuro Oncol.* 166 (2024) 351–357.
- [79] T. Jiang, D.H. Nam, Z. Ram, W.S. Poon, J. Wang, D. Boldbaatar, Y. Mao, W. Ma, Q. Mao, Y. You, C. Jiang, X. Yang, C. Kang, X. Qiu, W. Li, S. Li, L. Chen, X. Li, Z. Liu, W. Wang, H. Bai, Y. Yao, S. Li, A. Wu, K. Sai, G. Li, K. Yao, X. Wei, X. Liu, Z. Zhang, Y. Dai, S. Lv, L. Wang, Z. Lin, J. Dong, G. Xu, X. Ma, W. Zhang, C. Zhang, B. Chen, G. You, Y. Wang, Y. Wang, Z. Bao, P. Yang, X. Fan, X. Liu, Z. Zhao, Z. Wang, Y. Li, Z. Wang, G. Li, S. Fang, L. Li, Y. Liu, S. Liu, X. Shan, Y. Liu, R. Chai, H. Hu, J. Chen, W. Yan, J. Cai, H. Wang, L. Chen, Y. Yang, Y. Wang, L. Han, Q. Wang, G. Chinese, Glioma cooperative, C. Society for neuro-oncology of, A. Chinese brain cancer, A. Chinese glioma Genome, n. Asian glioma Genome Atlas, clinical practice guidelines for the management of adult diffuse gliomas, *Cancer Lett.* 499 (2021) 60–72.
- [80] T. Jiang, Y. Mao, W. Ma, Q. Mao, Y. You, X. Yang, C. Jiang, C. Kang, X. Li, L. Chen, X. Qiu, W. Wang, W. Li, Y. Yao, S. Li, S. Li, A. Wu, K. Sai, H. Bai, G. Li, B. Chen, K. Yao, X. Wei, X. Liu, Z. Zhang, Y. Dai, S. Lv, L. Wang, Z. Lin, J. Dong, G. Xu, X. Ma, J. Cai, W. Zhang, H. Wang, L. Chen, C. Zhang, P. Yang, W. Yan, Z. Liu, H. Hu, J. Chen, Y. Liu, Y. Yang, Z. Wang, Z. Wang, Y. Wang, G. You, L. Han, Z. Bao, Y. Liu, Y. Wang, X. Fan, S. Liu, X. Liu, Y. Wang, Q. Wang, CGCG clinical practice guidelines for the management of adult diffuse gliomas, *Cancer Lett.* 375 (2016) 263–273.
- [81] A. Choudhury, W.C. Chen, C.G. Lucas, J.C. Bayley, A.S. Harmanci, S.L.N. Maas, S. Santagata, T. Klisch, A. Perry, W.L. Bi, F. Sahm, A.J. Patel, S.T. Magill, D. R. Raleigh, Hypermutic meningiomas harbor DNA methylation subgroups with distinct biological and clinical features, *Neuro Oncol.* 25 (2023) 520–530.
- [82] D. Sturm, D. Capper, F. Andreiulio, M. Gessi, C. Kolsche, A. Reinhardt, P. Sievers, A.K. Wefers, A. Ebrahimi, A.K. Suwala, G.H. Gielen, M. Sill, D. Schimpf, D. Stichel, V. Hovestadt, B. Daenakas, A. Rode, S. Hamelmann, C. Previti, N. Jager, I. Buchhalter, M. Blattner-Johnson, B.C. Jones, M. Warmuth-Metz, B. Bison, K. Grund, C. Sutter, S. Hirsch, N. Dikow, M. Hasselblatt, U. Schuller, N. U. Gerber, C.L. White, M.K. Buntine, K. Kinross, E.M. Algar, J.R. Hansford, N. G. Gottardo, P. Hernaiz Driever, A. Gnekow, O. Witt, H.L. Muller, G. Calaminus, G. Fleischhack, U. Kordes, M. Mynarek, S. Rutkowski, M.C. Fruhwald, C. M. Kramm, A. von Deimling, T. Pietsch, F. Sahm, S.M. Pfister, D.T.W. Jones, Multimic neuropathology improves diagnostic accuracy in pediatric neuro-oncology, *Nat. Med.* 29 (4) (2023) 917–926.
- [83] T.A. Bale, M.K. Rosenblum, The 2021 WHO Classification of Tumors of the Central Nervous System: an update on pediatric low-grade gliomas and glioneuronal tumors, *Brain Pathol.* 32 (2022) e13060.
- [84] Y.-Z. Wang, Y.-W. Zhang, W.-H. Liu, R.-C. Chai, R. Cao, B. Wang, S.-Y. An, W.-J. Jiang, Y.-L. Xu, J. Yang, W.-Q. Jia, Spinal cord diffuse midline gliomas with H3 K27m-mutant: clinicopathological features and prognosis, *Neurosurgery* 89 (2021) 300–307.
- [85] L. Auffret, Y. Ajlil, A. Tauziède-Espariat, T. Kergrohen, C. Puiseux, L. Riffaud, P. Blouin, A.I. Bertozzi, P. Leblond, K. Blomgren, S. Froelich, A. Picca, M. Touat, M. Sanson, K. Beccaria, T. Blauwblomme, V. Dangouloff-Ros, N. Boddaert, P. Varlet, M.A. Debily, J. Grill, D. Castel, A new subtype of diffuse midline glioma, H3 K27 and BRAF/FGFR1 co-altered: a clinico-radiological and histomolecular characterisation, *Acta Neuropathol.* 147 (2023) 2.
- [86] Z. Zhao, K.-N. Zhang, Q. Wang, G. Li, F. Zeng, Y. Zhang, F. Wu, R. Chai, Z. Wang, C. Zhang, W. Zhang, Z. Bao, T. Jiang, Chinese glioma Genome Atlas (CGGA): a comprehensive resource with functional genomic data from Chinese glioma patients, *Dev. Reprod. Biol.* 19 (2021).
- [87] R. Yang, Y. Yu, Patient-derived organoids in translational oncology and drug screening, *Cancer Lett.* 562 (2023) 216180.
- [88] L. Qi, M. Kogiso, Y. Du, H. Zhang, F.K. Braun, Y. Huang, W.-Y. Teo, H. Lindsay, S. Zhao, P. Baxter, X. Zhao, L. Yu, Z. Liu, X. Zhang, J.M. Su, A. Adesina, J. Yang, M. Chintagumpala, L. Perlaky, C. Tsz-Kwong Man, C.C. Lau, X.-N. Li, Impact of SCID mouse gender on tumorigenicity, xenograft growth and drug-response in a large panel of orthotopic PDX models of pediatric brain tumors, *Cancer Lett.* 493 (2020) 197–206.
- [89] C.W. Mount, R.G. Majzner, S. Sundaresh, E.P. Arnold, M. Kadapakkam, S. Haile, L. Labanieh, E. Hulleman, P.J. Woo, S.P. Rietberg, H. Vogel, M. Monje, C. L. Mackall, Potent antitumor efficacy of anti-GD2 CAR T cells in H3-K27M+ diffuse midline gliomas, *Nat. Med.* 24 (2018) 572–579.
- [90] M. Filbin, M. Monje, Developmental origins and emerging therapeutic opportunities for childhood cancer, *Nat. Med.* 25 (2019) 367–376.
- [91] S. Jessa, A. Blanchet-Cohen, B. Krug, M. Vladouiu, M. Coutelier, D. Faury, B. Poreau, N. De Jay, S. Hebert, J. Monlong, W.T. Farmer, L.K. Donovan, Y. Hu, M.K. McConchy, F.M.G. Cavalli, L.G. Mikael, B. Ellezam, M. Richer, A. Allaire, A. G. Weil, J. Atkinson, J.P. Farmer, R.W.R. Dudley, V. Larouche, L. Crevier, S. Albrecht, M.G. Filbin, H. Sartelet, P.E. Lutz, C. Nagy, G. Turecki, S. Costantino, P.B. Dirks, K.K. Murai, G. Bourque, J. Ragoussis, L. Garzia, M.D. Taylor, N. Jabado, C.L. Kleinman, Stalled developmental programs at the root of pediatric brain tumors, *Nat. Genet.* 51 (2019) 1702–1713.
- [92] C.S. Grasso, Y. Tang, N. Truffaux, N.E. Berlow, L. Liu, M.A. Debily, M.J. Quist, L. E. Davis, E.C. Huang, P.J. Woo, A. Ponnuswami, S. Chen, T.B. Johung, W. Sun, M. Kogiso, Y. Du, L. Qi, Y. Huang, M. Hutt-Cabezas, K.E. Warren, L. Le Dret, P. S. Meltzer, H. Mao, M. Quezada, D.G. van Vuurden, J. Abraham, M. Fouladi, M. N. Svalina, N. Wang, C. Hawkins, J. Nazarian, M.M. Alonso, E.H. Raabe, E. Hulleman, P.T. Spellman, X.N. Li, C. Keller, R. Pal, J. Grill, M. Monje, Functionally defined therapeutic targets in diffuse intrinsic pontine glioma, *Nat. Med.* 21 (2015) 555–559.
- [93] R. Hashizume, N. Andor, Y. Ihara, R. Lerner, H. Gan, X. Chen, D. Fang, X. Huang, M.W. Tom, V. Ngo, D. Solomon, S. Mueller, P.L. Paris, Z. Zhang, C. Petrisch, N. Gupta, T.A. Waldman, C.D. James, Pharmacologic inhibition of histone demethylation as a therapy for pediatric brainstem glioma, *Nat. Med.* 20 (2014) 1394–1396.
- [94] A. Piunti, R. Hashizume, M.A. Morgan, E.T. Bartom, C.M. Horbinski, S. A. Marshall, E.J. Rendleman, Q. Ma, Y.H. Takahashi, A.R. Woodfin, A. V. Misharin, N.A. Abshiru, R.R. Lulla, A.M. Saratsis, N.L. Kelleher, C.D. James, A. Shilatfard, Therapeutic targeting of polycomb and BET bromodomain proteins in diffuse intrinsic pontine gliomas, *Nat. Med.* 23 (2017) 493–500.
- [95] D. Schapiro, H.W. Jackson, S. Raghuraman, J.R. Fischer, V.R.T. Zanotelli, D. Schulz, C. Giesen, R. Catena, Z. Varga, B. Bodenmiller, histoCAT: analysis of cell phenotypes and interactions in multiplex image cytometry data, *Nat. Methods* 14 (2017) 873–876.
- [96] B. Moran, M. Davern, J.V. Reynolds, N.E. Donlon, J. Lysaght, The impact of histone deacetylase inhibitors on immune cells and implications for cancer therapy, *Cancer Lett.* 559 (2023) 216121.
- [97] G.L. Lin, S. Nagaraja, M.G. Filbin, M.L. Suva, H. Vogel, M. Monje, Non-inflammatory tumor microenvironment of diffuse intrinsic pontine glioma, *Acta Neuropathol Commun* 6 (2018) 51.
- [98] N.A.P. Lieberman, K. DeGoller, H.M. Kovar, A. Davis, V. Hoglund, J. Stevens, C. Winter, G. Deutsch, S.N. Furlan, A.A. Vitanza, S.E. Leary, C.A. Crane, Characterization of the immune microenvironment of diffuse intrinsic pontine glioma: implications for development of immunotherapy, *Neuro Oncol.* 21 (2019) 83–94.
- [99] X. Fan, J. Li, B. Huang, H. Lu, C. Lu, M. Pan, X. Wang, H. Zhang, Y. You, X. Wang, Q. Wang, J. Zhang, Noninvasive radiomics model reveals macrophage infiltration in glioma, *Cancer Lett.* 573 (2023) 216380.
- [100] R.G. Majzner, S. Ramakrishna, K.W. Yeom, S. Patel, H. Chinnasamy, L.M. Schultz, R.M. Richards, L. Jiang, V. Barsan, R. Mancusi, A.C. Geraghty, Z. Good, A. Y. Mochizuki, S.M. Gillespie, A.M.S. Toland, J. Mahdi, A. Reschke, E.H. Nie, I. J. Chau, M.C. Rotiroti, C.W. Mount, C. Baggott, S. Mavroukakis, E. Egeler, J. Moon, C. Erickson, S. Green, M. Kunicki, M. Fujimoto, Z. Ehlinger, W. Reynolds, S. Kurra, K.E. Warren, S. Prabhu, H. Vogel, L. Rasmussen, T.

- T. Cornell, S. Partap, P.G. Fisher, C.J. Campen, M.G. Filbin, G. Grant, B. Sahaf, K. L. Davis, S.A. Feldman, C.L. Mackall, M. Monje, GD2-CAR T cell therapy for H3K27M-mutated diffuse midline gliomas, *Nature* 603 (2022) 934–941.
- [101] N.A. Vitanza, A.L. Wilson, W. Huang, K. Seidel, C. Brown, J.A. Gustafson, J. K. Yokoyama, A.J. Johnson, B.A. Baxter, R.W. Koning, A.N. Reid, M. Meechan, M. C. Biery, C. Myers, S.D. Rawlings-Rhea, C.M. Albert, S.R. Browd, J.S. Hauptman, A. Lee, J.G. Ojemann, M.E. Berens, M.D. Dun, J.B. Foster, E.E. Crotty, S.E.S. Leary, B.L. Cole, F.A. Perez, J.N. Wright, R.J. Orentas, T. Chour, E.W. Newell, J. R. Whiteaker, L. Zhao, A.G. Paulovich, N. Pinto, J. Gust, R.A. Gardner, M. C. Jensen, J.R. Park, Intraventricular B7-H3 CAR T cells for diffuse intrinsic pontine glioma: preliminary first-in-human bioactivity and safety, *Cancer Discov.* 13 (2023) 114–131.
- [102] N. Grassl, I. Poschke, K. Lindner, L. Bunse, I. Mildnerberger, T. Boschert, K. Jahne, E.W. Green, I. Hulsmeier, S. Junger, T. Kessler, A.K. Suwala, P. Eisele, M. O. Breckwoldt, P. Vajkoczy, O.M. Grauer, U. Herrlinger, J.C. Tonn, M. Denk, F. Sahm, M. Bendszus, A. von Deimling, F. Winkler, W. Wick, M. Platten, K. Sahm, A H3K27M-targeted vaccine in adults with diffuse midline glioma, *Nat. Med.* 29 (2023) 2586–2592.
- [103] I. Ausejo-Mauleon, S. Labiano, D. de la Nava, V. Laspidea, M. Zalacain, L. Marrodan, M. Garcia-Moure, M. Gonzalez-Huarriz, I. Hervás-Corpion, L. Dhandapani, S. Vicent, M. Collantes, I. Penuelas, O.J. Becher, M.G. Filbin, L. Jiang, J. Labelle, C.A.O. de Biagi-Junior, J. Nazarian, S. Laternser, T. N. Phoenix, J. van der Lugt, M. Kranendonk, R. Hoogendijk, S. Mueller, C. De Andrea, A.C. Anderson, E. Guruceaga, C. Koschmann, V.N. Yadav, J. Gallego Perez-Larraya, A. Patino-Garcia, F. Pastor, M.M. Alonso, TIM-3 blockade in diffuse intrinsic pontine glioma models promotes tumor regression and antitumor immune memory, *Cancer Cell* 41 (2023) 1911–1926 e1918.
- [104] J. Bahney, C.S. von Bartheld, The cellular composition and glia-neuron ratio in the spinal cord of a human and a nonhuman primate: comparison with other species and brain regions, *Anat. Rec.* 301 (2018) 697–710.
- [105] D.F. Quail, J.A. Joyce, The microenvironmental landscape of brain tumors, *Cancer Cell* 31 (2017) 326–341.
- [106] Q. Zhang, S. Cheng, Y. Wang, M. Wang, Y. Lu, Z. Wen, Y. Ge, Q. Ma, Y. Chen, Y. Zhang, R. Cao, M. Li, W. Liu, B. Wang, Q. Wu, W. Jia, X. Wang, Interrogation of the microenvironmental landscape in spinal ependymomas reveals dual functions of tumor-associated macrophages, *Nat. Commun.* 12 (2021) 6867.
- [107] J.A. Ellis, M. Castelli, M. Assanah, J.N. Bruce, P. Canoll, A.T. Ogden, Unique microenvironmental responses to PDGF stimulation in brain and spinal cord gliomas determine tumor phenotype, *J. Neuro Oncol.* 123 (2015) 27–33.
- [108] M.C. Chamberlain, S.K. Johnston, Recurrent spinal cord glioblastoma: salvage therapy with bevacizumab, *J. Neuro Oncol.* 102 (2011) 427–432.
- [109] K. Madhavan, I. Balakrishnan, S. Lakshmanachetty, A. Pierce, B. Sanford, S. Fosmire, H.B. Elajaili, F. Walker, D. Wang, E.S. Nozik, S.S. Mitra, N.A. Dahl, R. Vibhakar, S. Venkataraman, Venetoclax cooperates with ionizing radiation to attenuate diffuse midline glioma tumor growth, *Clin. Cancer Res.* 28 (2022) 2409–2424.
- [110] Q.T. Storozynsky, K.C. Agopsowicz, R.S. Noyce, A.B. Bukhari, X. Han, N. Snyder, B.A. Umer, A.M. Gamper, R. Godbout, D.H. Evans, M.M. Hitt, Radiation combined with oncolytic vaccinia virus provides pronounced antitumor efficacy and induces immune protection in an aggressive glioblastoma model, *Cancer Lett.* 562 (2023) 216169.
- [111] G. Chen, D. Zheng, Y. Zhou, S. Du, Z. Zeng, Olaparib enhances radiation-induced systemic anti-tumor effects via activating STING-chemokine signaling in hepatocellular carcinoma, *Cancer Lett.* 582 (2024) 216507.
- [112] C.C. Pinnix, L. Chi, E.J. Jabbour, S.A. Milgrom, G.L. Smith, N. Daver, N. Garg, M. D. Cykowski, G. Fuller, D. Cachia, C. Kamiya-Matsuoka, K. Woodman, C. Dinardo, N. Jain, T.M. Kadia, N. Pemmaraju, M. Ohanian, M. Konopleva, H.M. Kantarjian, B.S. Dabaja, Dorsal column myelopathy after intrathecal chemotherapy for leukemia, *Am. J. Hematol.* 92 (2017) 155–160.
- [113] S.D. Kaur, N. Bedi, D. Kumar, D.N. Kapoor, PROTACs: promising approach for anticancer therapy, *Cancer Lett.* 556 (2023) 216065.